Degrees of Freedom and Range of Motion



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Actuators Outputs

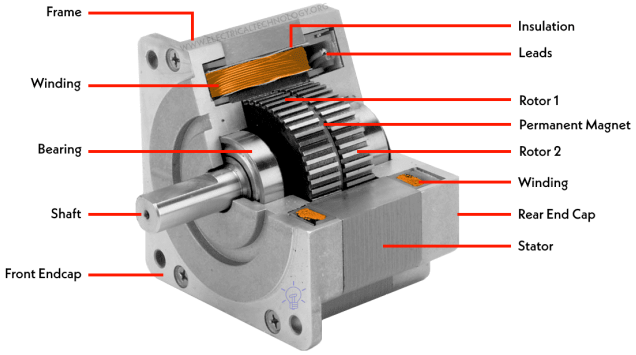
- **Linear Actuators**: Produce straight-line motion (e.g., hydraulic pistons, lead screw actuators).
- **Rotary Actuators**: Produce rotational motion (e.g., motors, rotary solenoids).
- **Combination Actuators**: Use mechanisms like cams or gears to combine motions.





Electric Actuators - Motors

- **Purpose:** Convert electrical energy into mechanical movement (rotation or motion).
- **How they work:** When an electric current flows through a coil inside a magnetic field, it produces a force (the motor effect) that makes the shaft turn.
- **Common Types:**
  - DC Motors – simple, rotate when powered; speed changes with voltage.
  - Stepper Motors – rotate in precise steps, useful for positioning.
  - Servo Motors – allow controlled angle or position movement.
- **What happens when voltage increases:**
  - More current flows → motor spins faster or with more torque (depending on type).



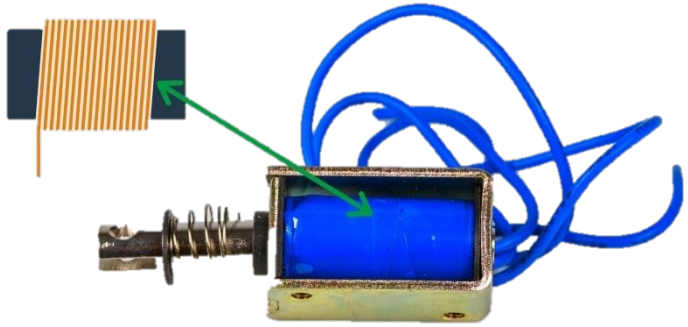
A stepper motor
Output – Rotary
Input – Power & Control Signal



An electric motor
Output – Rotary
Input – Power

Electric Actuators - Solenoid

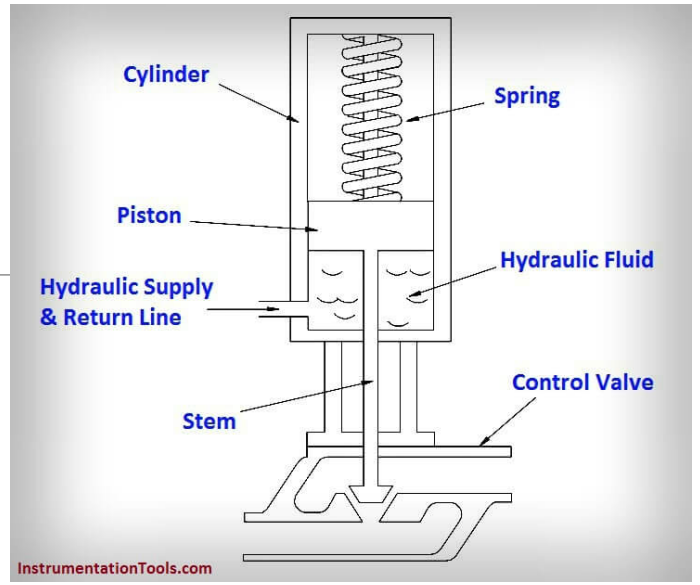
- **Purpose:** Convert electrical energy into a linear (straight-line) pushing or pulling motion.
- **How they work:** A coil of wire creates a magnetic field when current flows. This magnetic field pulls or pushes a metal rod (plunger) inside the coil.
- **What happens when voltage is applied:**
  - Current flows through the coil → magnetic field is created → plunger moves in or out.
- **Types of Solenoids:**
  - Pull-type – plunger is pulled into the coil.
  - Push-type – plunger is pushed out when activated.



An electric solenoid
Output – Linear
Input – Power

Hydraulic Actuators

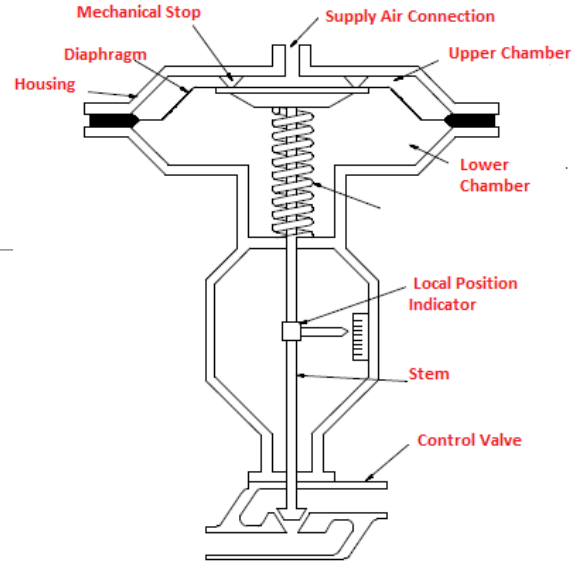
- **Purpose:** Use pressurised liquid (usually oil) to create linear or rotary movement with high force.
- **How they work:** A pump forces fluid into a cylinder or motor. The fluid pressure pushes against a piston (linear motion) or rotates a shaft (rotary motion).
- **What happens when pressure increases:**
  - More fluid pressure → greater force or torque output.
- **Types of Hydraulic Actuators:**
  - Hydraulic Cylinder – produces straight-line (linear) movement.
  - Hydraulic Motor – produces rotary motion.

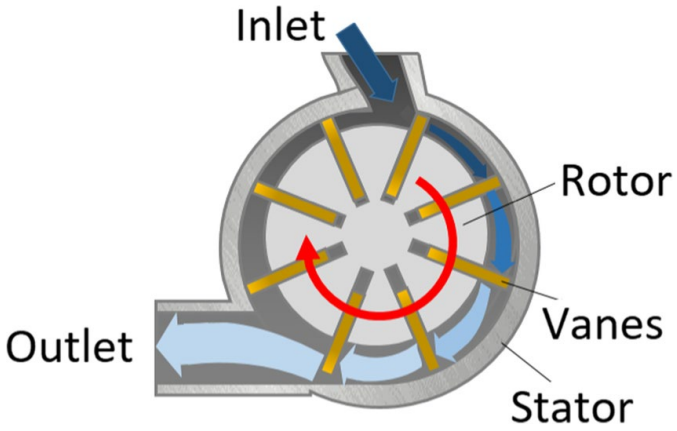




Pneumatic Actuators

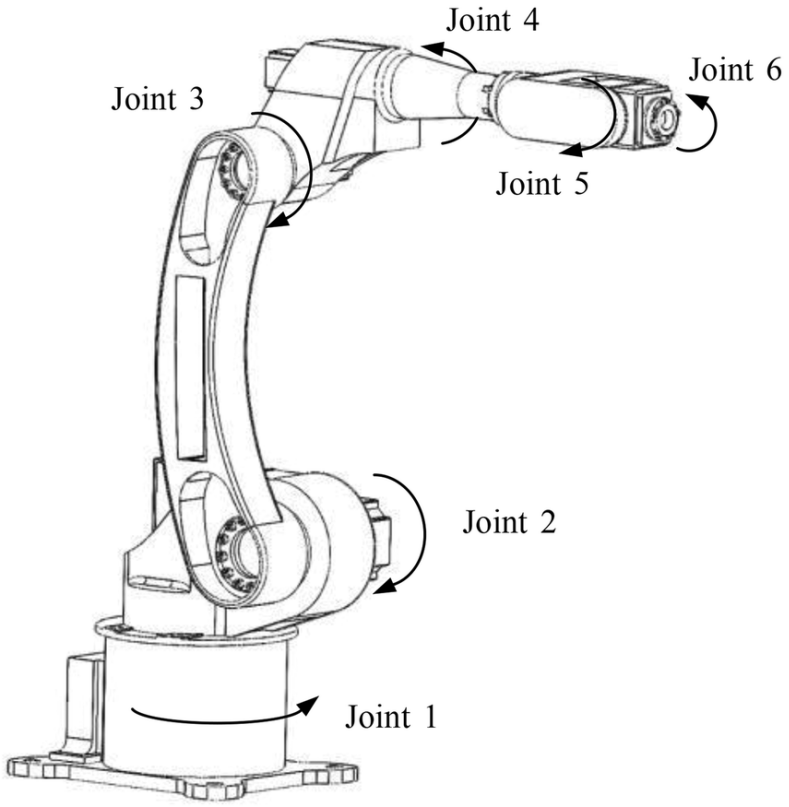
- **Purpose:** Use compressed air to create linear or rotary motion.
- **How they work:** Air pressure is applied to a piston inside a cylinder (for linear motion) or to a vane/rotor (for rotary motion). The expansion of compressed air pushes the actuator into movement.
- **What happens when air pressure increases:**
  - Higher pressure → stronger movement or force output.
- **Types of Pneumatic Actuators:**
  - Cylinders – provide straight-line pushing or pulling.
  - Pneumatic motors/rotary actuators – provide rotation.





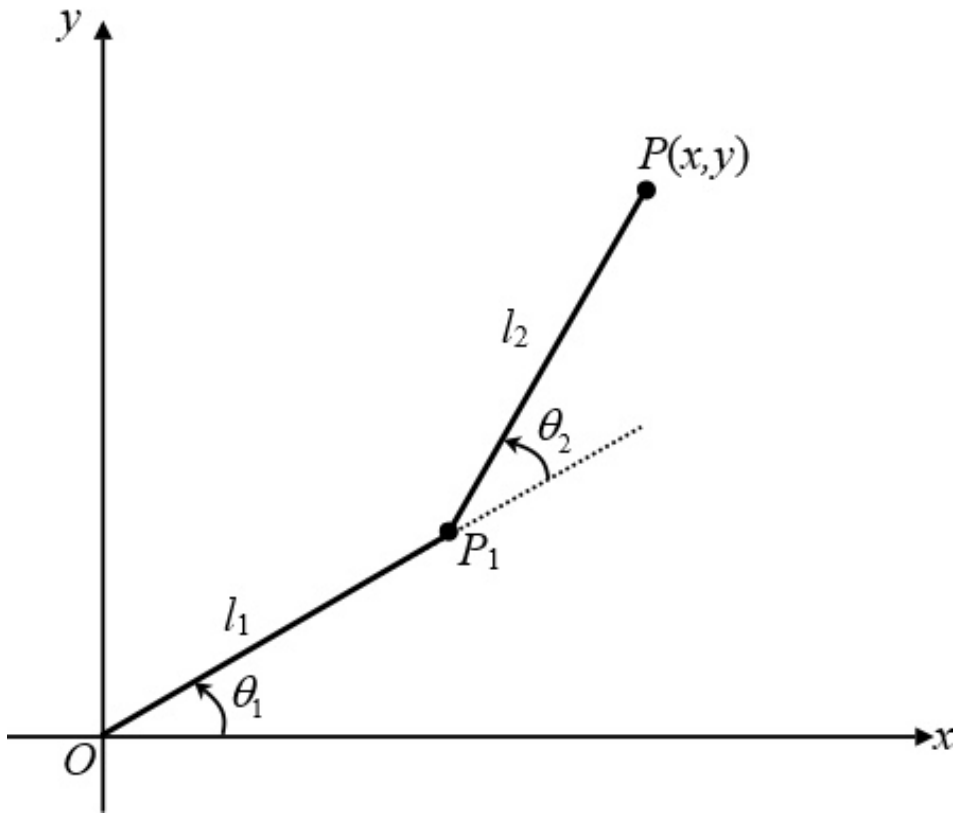
Degrees of Freedom (DoF)

- **Definition**: *The number of independent ways a robot or mechanism can move.*
- *Each DoF = one type of motion (translation or rotation).*
- *More DoF = more flexibility and complexity.*



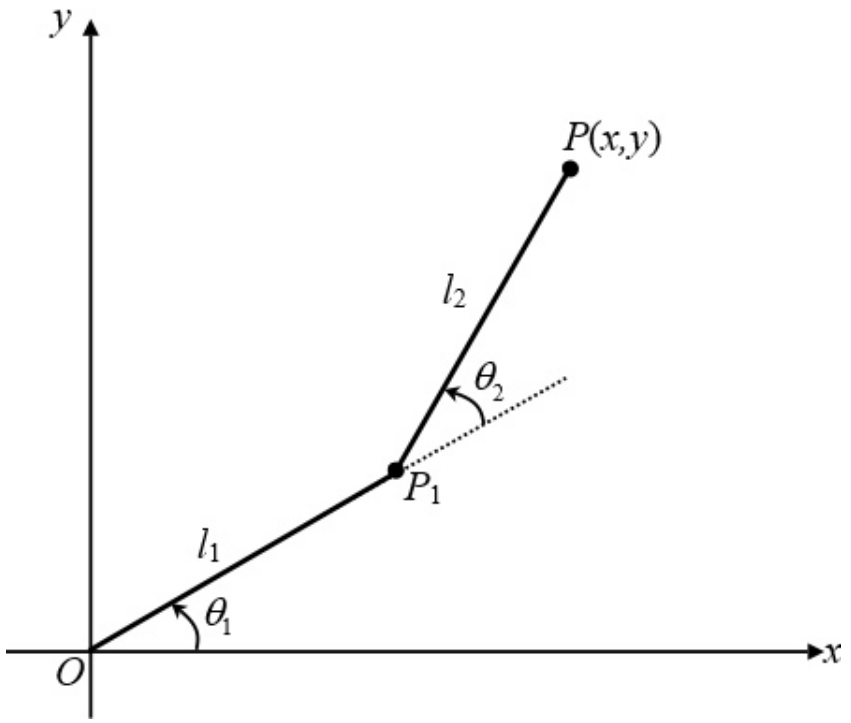
2D Motion

- To understand DoF we will start with 2D motion
- In 2D motion we have 3 possible motions:
  - Move around increasing or decreasing x and y
  - Rotate on a fixed point



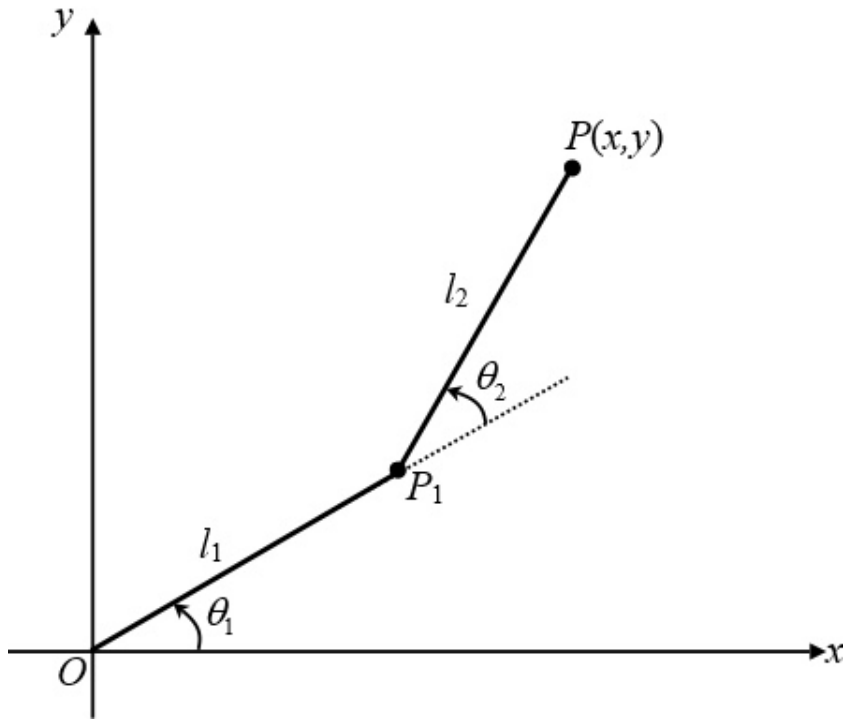
2D Motion

- So, for a robot arm which only exists in 2d space its joints can either:
  - Extend and retract linearly changing x or y
  - Rotate around a fixed-point changing x and y of a different part of the arm



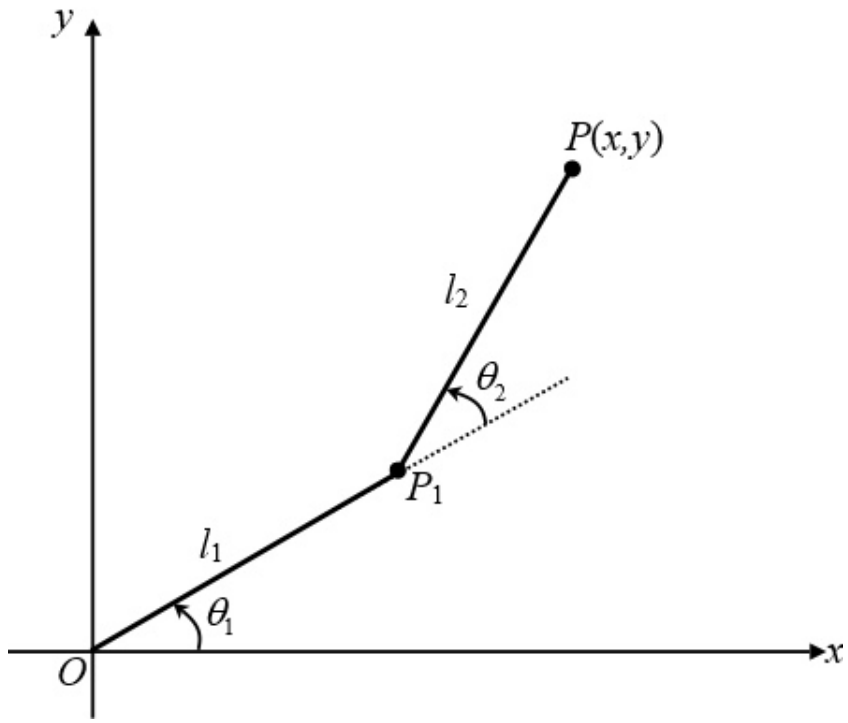
2D range of motion

- So now we can explain the possible ways the whole arm can move
- Each joint in our 2D arm has a limited range of movement:
  - A limited extension length for a piston in x and y
  - A limited rotation based on motor limits or collision with other parts of the arm



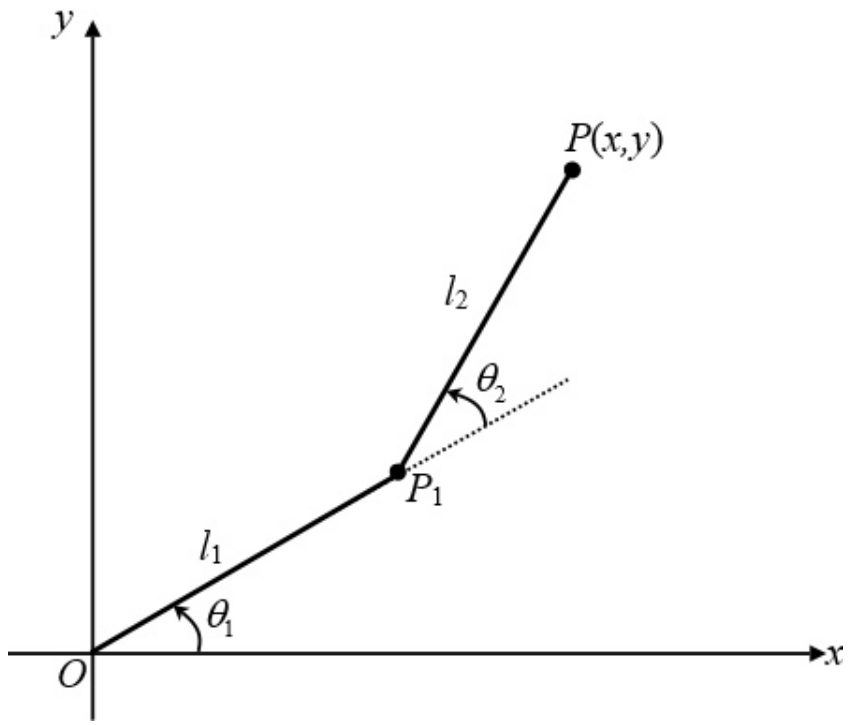
2D Degrees of Freedom

- So, for this 2D arm we can simplify it drastically into a collection of joints
- And for each joint which makes up the arm we can take its motion and record it as a degree of freedom
- Typically, one joint = one DoF (but not always; some joints allow more than one type of movement).



2D Degrees of Freedom

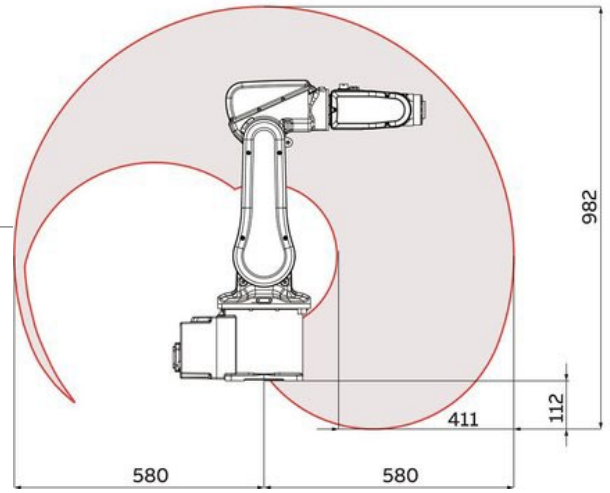
- The arm on this slide has 2 joints (θ_1 and θ_2)
- Each of these joints only has rotational motion
- So, the arm has 2 joints, each with 1 motion
- This means the arm has 2 DoF

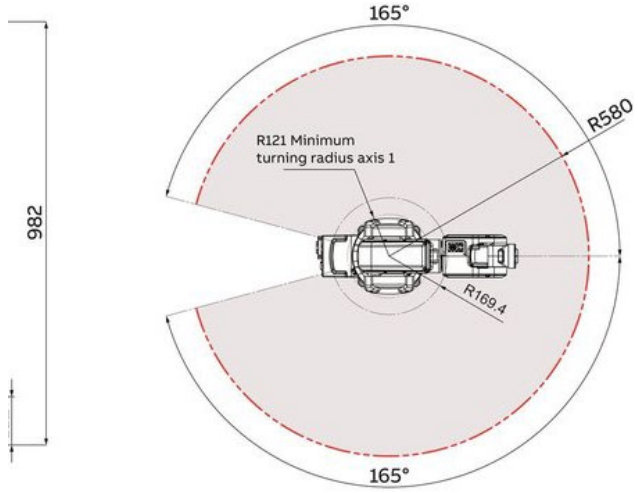


3D RoM and DoF

- 3D range of motion works the same as 2D, we take each joint or actuator and look at how it moves and how much it can move

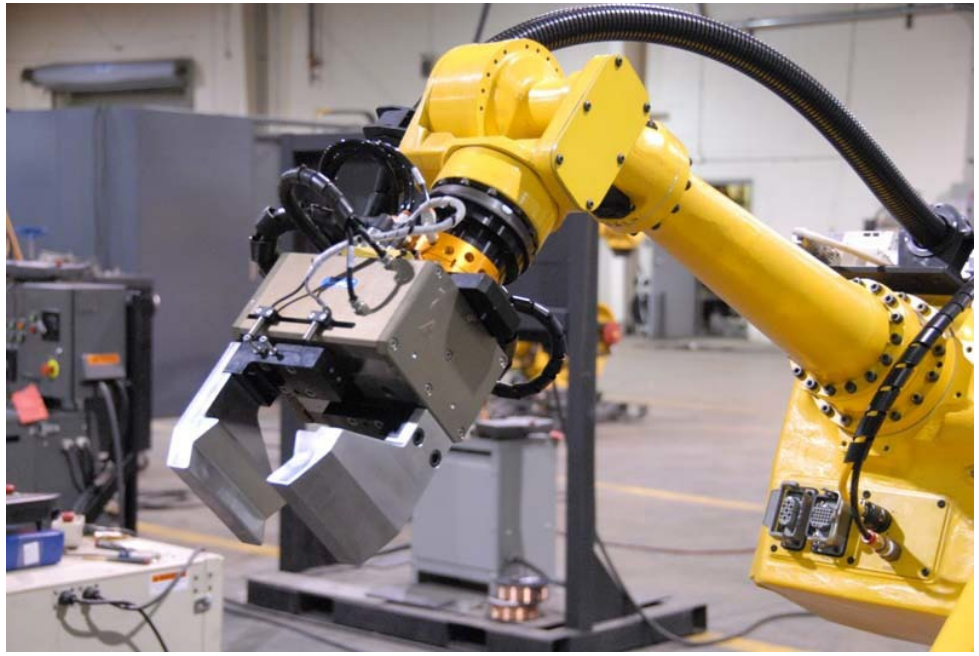
- Often this involves a large bottom actuator which rotates our whole 2D system





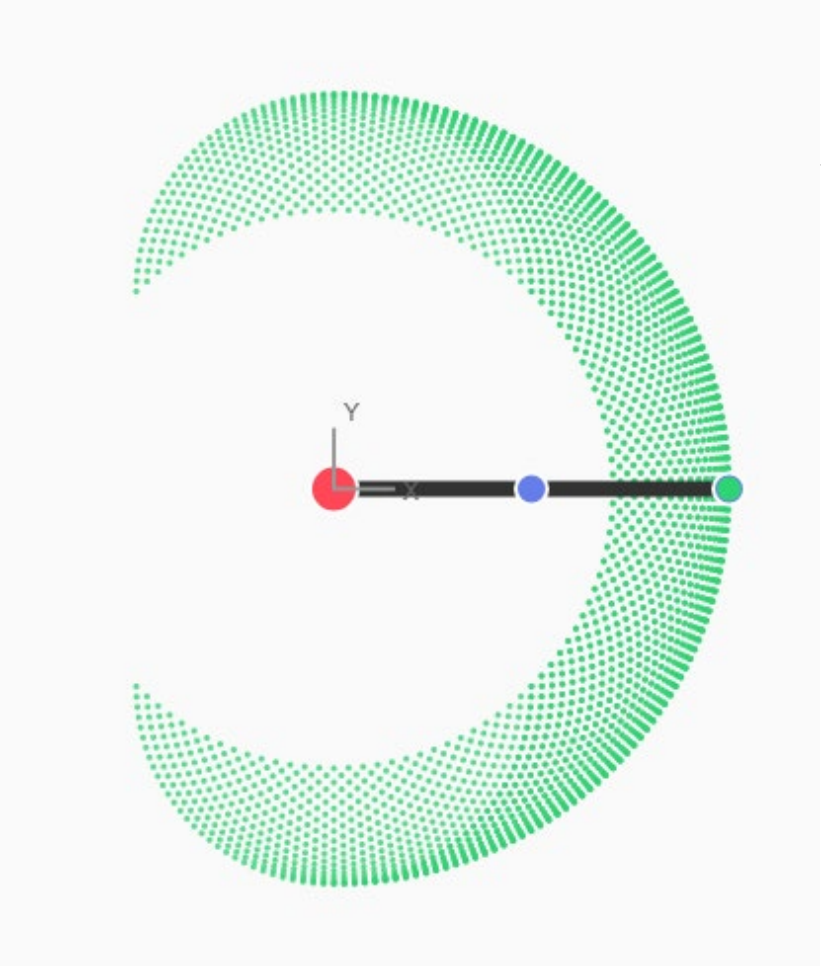
End Effector

- The end effector is the final point in our robot arm.
- It's often the point we'd attach a gripper or an actuator to do a task
- It's the part of the arm that does the action



2D workspace

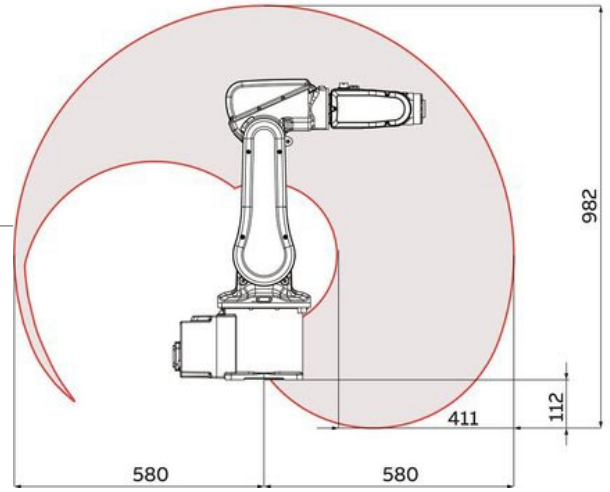
- Our DoF and effect the workspace our robot arm can reach
- This workspace is the area that the end effector can reach
- Its important to know as it determines how you'd use the robot arm

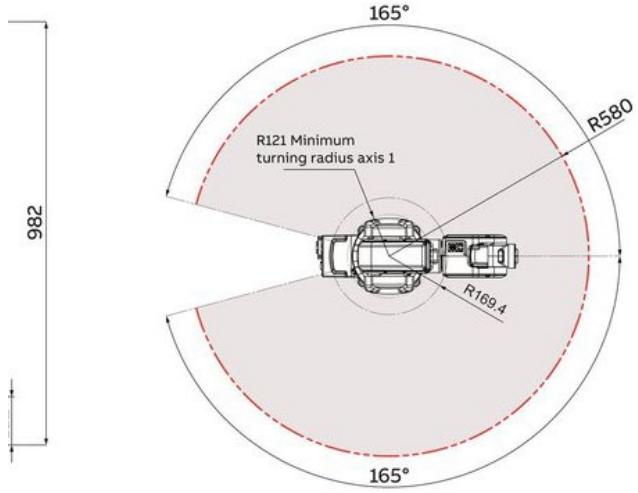


3D workspace

- We can expand our 2D workspace and rotate it around a point making a continuous shape

- This shape is the point the end effector of the arm can reach



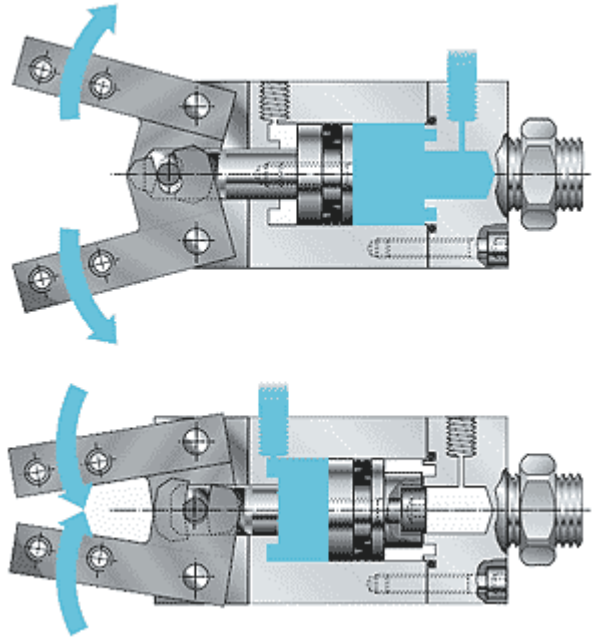


Types of end effector - Pneumatic

- Definition: End effectors that use compressed air to grip, move, or release objects.

- Common types:
  - Suction cups (vacuum grip)
  - Pneumatic grippers (two-jaw or three-jaw)
  - Blowing/air-jet systems

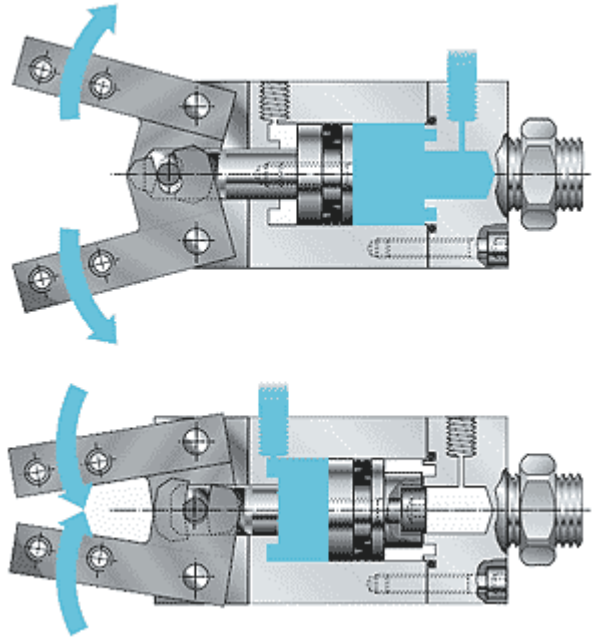




Types of end effector - Pneumatic

- **How they work:**
  - Powered by a pneumatic system (compressed air supply + valves + cylinders).
  - Air pressure generates motion or suction.
  - Controlled via solenoid valves or PLCs.
  - Can be designed for pick-and-place, packaging, or assembly tasks.

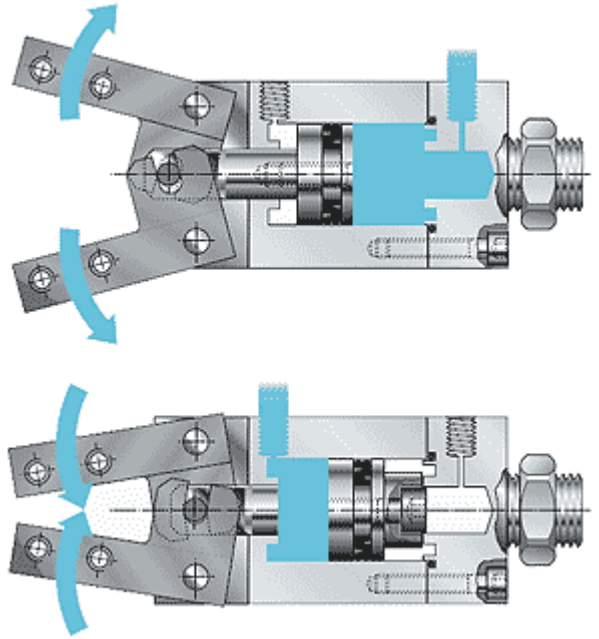




Types of end effector - Pneumatic

- **Advantages**
  - Lightweight and simple design.
  - Fast operation – quick open/close.
  - Cost-effective compared to hydraulic systems.
  - Safe for delicate items (no sharp edges).
  - Easy to integrate with automation systems

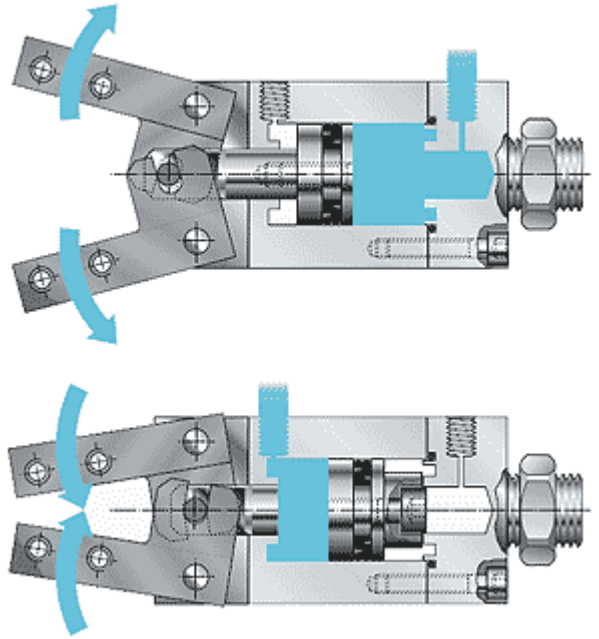




Types of end effector - Pneumatic

- **Disdvantages**
  - Limited gripping force – not suitable for very heavy loads.
  - Requires constant air supply and good sealing.
  - Less precise than electric or hydraulic alternatives.
  - Can be noisy (air release)..





Types of end effector - Electric

- Definition: End effectors powered by electric motors, solenoids, or actuators.
- Provide precise, programmable control over gripping and manipulation.
- Common types:
  - Servo-driven grippers
  - Solenoid-based tools
  - Electric screwdrivers, welders, or cutters



Types of end effector - Electric

- Use electric motors or actuators to generate motion.
- Controlled directly by the robot’s controller or PLC.
- Can provide position feedback using encoders or sensors.
- Allow fine control of grip force, speed, and position.



Types of end effector - Electric

- Use electric motors or actuators to generate motion.
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Types of end effector - Electric

- **Advantages**
  - High precision and repeatability.
  - Adjustable grip strength for different objects.
  - Can handle complex or delicate tasks.
  - Lower maintenance than pneumatic/hydraulic systems.
  - Easy integration with sensors and feedback systems..



Types of end effector - Electric

- **Disadvantages**
  - More expensive than simple pneumatic grippers.
  - Typically slower than pneumatic systems for very rapid cycles.
  - Less force compared to hydraulic end effectors.
  - Requires electrical wiring and drivers.

